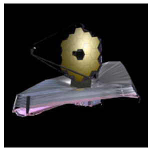

7965 - Probing the chemical evolution of the carriers of the Aromatic Infrared Band emission.

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1	NIRSpec IFU	NIRSpec IFU Spectroscopy	(1) NGC-6644
	2	NIRSpec IFU sky	NIRSpec IFU Spectroscopy	(2) NGC-6644-BACKGROUND
	3	MIRI MRS	MIRI Medium Resolution Spectroscopy	(1) NGC-6644
	4	MIRI MRS sky	MIRI Medium Resolution Spectroscopy	(2) NGC-6644-BACKGROUND

ABSTRACT

The IR spectra of regions associated with gas and dust often exhibit strong emission features known as the Aromatic Infrared Bands (AIBs). In circumstellar environments of both young stellar objects and old stars, the AIBs display clear spectral variations that reflect changes in their chemical make up, more specifically in the aromaticity of the AIB carriers. The aromaticity of the AIB carriers has a profound influence on key physical processes such as the photo-electric effect and ionization. This in turn sets the thermal and ionization balance in planet-forming disks around young stellar objects (among others) which may influence the formation of planetesimals and planets.

Here, we propose 0.97-28 micron NIRSpec IFU and MIRI-MRS spatially resolved observations of the planetary nebula NGC 6644 to characterize the spatial behaviour of the AIB emission in its circumstellar toroid. The goal of the program is to study the processes that regulate the aliphatic-to-aromatic character of the AIB carriers as well as the relationship between the degree of aromaticity and the spectral characteristics of the AIB emission. We aim to establish an evolutionary link between the highly aromatic AIBs typically seen in the ISM (class A) and the more aliphatic AIBs seen in circumstellar environments (class B), and determine the conditions that promote this evolution.

OBSERVING DESCRIPTION

This program includes observations with 1) NIRSpec IFU using the G140H/F100LP, F235H/F170LP, G395H/F290LP grating/filter combinations and a 4-point dither pattern and 2) MIRI MRS in all channels using a 4-point dither. The observations have a dedicated NIRSpec leakcal observation and dedicated sky observations. We perform MIRI/MRS simultaneous imaging.

The NIRSpec and MRS observations are linked as an uninterruptible sequence, but otherwise the observation are unconstrained.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-6644	RA: 18 32 34.6970 (278.1445708d) Dec: -25 07 44.11 (-25.12892d) Equinox: J2000	Epoch of Position: 2000	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Star Description=[Planetary nebulae nuclei] Extended=YES				
(2)	NGC-6644-BACKGROUND	RA: 18 32 35.4330 (278.1476375d) Dec: -25 07 13.69 (-25.12047d) Equinox: J2000	Epoch of Position: 2015.5	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Calibration Description=[Telescope/sky background] Extended=YES				

Observation

Proposal 7965, Observation 1: NIRSpec IFU

Tue Jul 22 07:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observations:[NIRSpec IFU sky (Obs 2)]

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-6644	RA: 18 32 34.6970 (278.1445708d) Dec: -25 07 44.11 (-25.12892d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Planetary nebulae nuclei]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	NRSIRS2RAPID	5	12	false	true	NONE	4	48	4201.6	227283
2	G395H/F290LP	NRSIRS2RAPID	5	12	true	true	NONE	4	48	4201.6	227283
3	G140H/F100LP	NRSIRS2RAPID	5	2	false	true	NONE	4	8	700.267	227283
4	G140H/F100LP	NRSIRS2RAPID	5	2	true	true	NONE	4	8	700.267	227283
5	G235H/F170LP	NRSIRS2RAPID	5	2	false	true	NONE	4	8	700.267	227283
6	G235H/F170LP	NRSIRS2RAPID	5	2	true	true	NONE	4	8	700.267	227283

Special Requirements	Group Observations 1, 2, Non-interruptible

Observation

Proposal 7965, Observation 2: NIRSpec IFU sky Tue Jul 22 07:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec IFU Spectroscopy

Background Observation For: [NIRSpec IFU (Obs 1)]

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	NGC-6644-BACKGROUND	RA: 18 32 35.4330 (278.1476375d) Dec: -25 07 13.69 (-25.12047d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

Template

TA Method	HFF Readout Mode
NONE	false

Dithers

#	Dither Type	Size	Starting Point	Number of Points	Points
1	4-POINT-DITHER				

Spectral Elements

#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G395H/F290LP	NRSIRS2RAPID	5	12	false	true	NONE	4	48	4201.6	
2	G140H/F100LP	NRSIRS2RAPID	5	2	false	true	NONE	4	8	700.267	
3	G235H/F170LP	NRSIRS2RAPID	5	2	false	true	NONE	4	8	700.267	

Special Requirements

Group Observations 1, 2, Non-interruptible

Observation

Proposal 7965, Observation 3: MIRI MRS

Tue Jul 22 07:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observations:[MIRI MRS sky (Obs 4)]

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-6644	RA: 18 32 34.6970 (278.1445708d) Dec: -25 07 44.11 (-25.12892d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Star
Description=[Planetary nebulae nuclei]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
F560W	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F1130W	FASTR1	23	1	1	Dither 1	4	4	255.304	
1	LONG(C)	MRSLONG		FASTR1	5	30	1	Dither 1	4	120	1986.929	227283
1	LONG(C)	MRSSHORT		FASTR1	5	30	1	Dither 1	4	120	1986.929	227283
2		IMAGER	F1000W	FASTR1	23	1	1	Dither 1	4	4	255.304	
2	MEDIUM(B)	MRSLONG		FASTR1	5	30	1	Dither 1	4	120	1986.929	227283
2	MEDIUM(B)	MRSSHORT		FASTR1	5	30	1	Dither 1	4	120	1986.929	227283
3		IMAGER	F770W	FASTR1	23	1	1	Dither 1	4	4	255.304	
3	SHORT(A)	MRSLONG		FASTR1	10	16	1	Dither 1	4	64	1942.528	227283
3	SHORT(A)	MRSSHORT		FASTR1	10	16	1	Dither 1	4	64	1942.528	227283

Special Requirements	Sequence Observations 3, 4, Non-interruptible

Observation

Proposal 7965, Observation 4: MIRI MRS sky

Tue Jul 22 07:00:11 GMT 2025

Diagnostic Status: Warning

Observing Template: MIRI Medium Resolution Spectroscopy

Background Observation For: [MIRI MRS (Obs 3)]

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	NGC-6644-BACKGROUND	RA: 18 32 35.4330 (278.1476375d) Dec: -25 07 13.69 (-25.12047d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Calibration
Description=[Telescope/sky background]
Extended=YES

Acquisition

#	Target
1	NONE

Template

AcqFilter	Primary Channel	Simultaneous Imaging	Imager Subarray	Grating Wheel Direction
	All MRS	YES	FULL	Allow Auto Reorder

Dithers

#	Dither Type	Optimized For	Direction
1	4-Point	EXTENDED SOURCE	NEGATIVE

Spectral Elements

#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1		IMAGER	F1130W	FASTR1	5	1	1	Dither 1	4	4	55.501	
1	LONG(C)	MRSLONG		FASTR1	5	30	1	Dither 1	4	120	1986.929	
1	LONG(C)	MRSSHORT		FASTR1	5	30	1	Dither 1	4	120	1986.929	
2		IMAGER	F1000W	FASTR1	5	1	1	Dither 1	4	4	55.501	
2	MEDIUM(B)	MRSLONG		FASTR1	5	30	1	Dither 1	4	120	1986.929	
2	MEDIUM(B)	MRSSHORT		FASTR1	5	30	1	Dither 1	4	120	1986.929	
3		IMAGER	F770W	FASTR1	5	1	1	Dither 1	4	4	55.501	
3	SHORT(A)	MRSLONG		FASTR1	10	16	1	Dither 1	4	64	1942.528	
3	SHORT(A)	MRSSHORT		FASTR1	10	16	1	Dither 1	4	64	1942.528	

Special Requirements	Sequence Observations 3, 4, Non-interruptible